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Model Set -1

Applied Chemistry-A

Group-A

1. Choose the most suitable answer from the following options:

(i) The number of radial nodes for 3p orbital is:

- (a) 1 (b) 2 (c) 3 (d) 4

Ans.(a)

(ii) Ratio of distance between two electrodes and area of the electrodes is called

- (a) Planks constant (b) Cell constant
(c) Gas constant (d) Rydberg's constant

Ans.(b)

(iii) Biochemical Oxygen Demand (BOD) is a measure of organic material present in water, BOD value less than 5ppm indicates a water sample to be:-

- (a) Rich in dissolved oxygen
(b) Poor in dissolved oxygen
(c) Highly polluted
(d) Not suitable for aquatic life

Ans.(a)

(iv) Total number of orbitals associated with third shell will be :

- (a) 2 (b) 4 (c) 9 (d) 3

Ans.(c)

(v) The presence of five unpaired electron in a d orbital can be explained by-

- (a) Pauli's exclusion principle
(b) Aufbau principle
(c) Hund's rule
(d) None of these

Ans.(c)

(vi) Protium, Deuterium and Tritium are of hydrogen:-

- (a) Isotopes (b) Isotones
(c) Isomers (d) None of these

Ans.(a)

(vii) Total number of orbital associated with second shell will be:

- (a) 2 (b) 3 (c) 4 (d) 9

Ans.(b)

(viii) What is the following polymer is thermoplastic?

- (a) Teflon (b) Bakelite
(c) Natural rubber (d) Neoprene

Ans.(a)

(ix) The unit of cell constant is :-

- (a) $\text{Ohm}^{-1} \text{cm}^{-1}$ (b) Ohm cm (c) cm (d) cm^{-1}

Ans.(d)

(x) Which of the following is not used to construct salt bridge?

- (a) KCl (b) KNO_3 (c) NH_4NO_3 (d) CH_3COOK

Ans.(d)

(xi) Mica is an ore of:-

- (a) Fe (b) Al (c) Cu (d) Cr

Ans.(b)

(xii) The difference between the electrode potentials of two electrodes when no current is drawn through the cell is called:-

- (a) Cell constant (b) Cell emf
(c) Potential difference (d) Cell voltage

Ans.(b)

(xiii) Which of the following polymer is an example of condensation polymer?

- (a) PVC (b) Buna-S (c) Butyl rubber (d) Bakelite

Ans.(b)

(xiv) Fused NaCl conducts electricity because of :

- (a) Free ions (b) Free electrons
(c) Free molecules (d) None of these

Ans.(a)

(xv) In Daniel cell, when Zn and Cu electrodes are connected the electron flows from :

- (a) Cu to Zn outside the cell
(b) Zn to Cu outside the cell
(c) Electrolyte to Zn electrode
(d) None of these

Ans.(a)

(xvi) Sulphide Ores are concentrated by

- (a) Hydraulic washing (b) Gravity separation
(c) Electromagnetic separation
(d) Froth flotation process

Ans.(d)

(xvii) For reducing 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} the charge required is:-

- (a) 96500 C (b) $3 \times 96500 \text{ C}$
(c) $6 \times 96500 \text{ C}$ (d) None of these

Ans.(b)

(xviii) Which one of the following can have only negative oxidation state?

- (a) F (b) I (c) Na (d) Ne

Ans.(a)

(xix) Which element is not obtained by electrolytic process?

- (a) Na (b) Fe (c) Ca (d) None

Ans.(c)

(xx) $\text{pH} + \text{pOH}$ of neutral water is

- (a) 7 (b) 10 (c) 14 (d) 3

Ans.(c)

Group-B

Answer the following Questions:

Q2. Explain Pauli's exclusion principle.

Ans. Refer to Chapter 1 of Venus Publication Question Bank

OR

Q2. Explain the variation of molar conductance with concentration.

Ans. Variation of molar conductance with concentration:

The molar conductivity is the conductance of all the ions produced by one mole of electrolyte. On dilution the total number of ions increases due to increase in the degree of dissociation. Hence the molar conductivity of both strong and weak electrolytes increases with dilution.

The relation between conductivity and molar conductivity is given by $\Lambda = \frac{k}{C} = k \cdot V$. On dilution although k

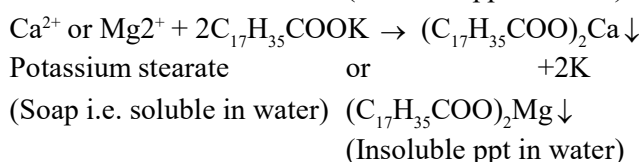
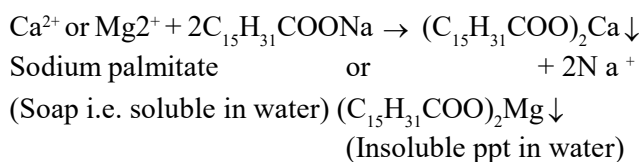
decreases, increase in volume V is more than compensated by the decrease in value of k .

On dilution, the molar conductivity of strong electrolytes increases upto the maximum limiting value Λ_0 i.e. value at zero concentration or at ∞ dilution. The molar conductivity reaches the value Λ_0 in just 0.001 or 0.0001M solutions of strong electrolytes but for weak electrolytes at these concentrations it is less than Λ_0 .

Q3. What are the causes of hardness of water ?

Ans. Water contains many dissolved salts when it passes over rocks and through the various underlying layers of earth. Bicarbonates, chlorides and sulphates of calcium and magnesium are the common salts which dissolve in water. Water containing any of these salts does not produce lather with soap easily and is explained as follows:

Soaps are the potassium and sodium salts of higher fatty acids like stearic acid ($C_{17}H_{35}COOH$), palmitic acid ($C_{15}H_{31}COOH$) etc. When hard water containing Ca^{2+} or Mg^{2+} ions reacts with a soap solution, a precipitate of the calcium magnesium salts of the respective fatty acid is formed as below:



Hence, when we use hard water, soap gets precipitated in the form of insoluble calcium and magnesium salts. Therefore, no lather is produced till all the calcium or magnesium ions get precipitated and more amount of soap is required for the formation of lather. So, hard water is not useful for the washing purposes.

OR

Q3. Why platinum, gold and silver are used in making jewellery?

Ans: Platinum, gold and silver are used in making jewellery because metals have lustre, and these are less reactive, these metals cannot react with air, water and acids, and these metals are malleable and ductile. Due to these properties these metals are called noble metals and they are used in jewellery.

Q4. What is the difference between potential difference and electromotive force?

Ans. Refer to Chapter 5 of Venus Publication Question Bank

OR

Q4. Find the weight of copper deposited from copper sulphate solution by a current of 0.25 ampere flowing for one hour, {Cu=63.5}.

Ans: Current = 0.25 ampere

Time = 1 hr = 1 x 60 x 60 = 3600 sec.

$$\begin{aligned} \therefore \text{Quantity of electricity passed} &= \text{Current} \times \text{Time} \\ &= 0.25 \times 3600 \\ &= 900C \end{aligned}$$

Assume the reaction occurring at the cathode is



Thus, 2F i.e. 2 x 96500 C deposit Cu = 1 mol = 63.5g.

$\therefore$ 900C will deposit

$$Cu = \frac{63.5}{2 \times 96500} \times 900 = 0.296g$$

Q5. An element with mass number 81 contains 31.7% more neutrons than protons. What is the symbol and atomic number?

Ans. $p + n = 81$ (i)

$$n = \frac{31.7}{100} \times p + p$$

$$n = 1.317p$$
(ii)

Putting equation (ii) in equation (i)

$$p + 1.317p = 81$$

$$2.317p = 81$$

$$p = \frac{81}{2.317} = 34.95 \approx 35$$

Since p has to be a whole number, $p = 35$

$$\therefore n = 81 - 35 = 46$$

Hence, the symbol is $^{81}_{35}Br$

OR

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Q5. Discuss the purpose of vulcanisation of rubber.

Ans. Refer to Chapter 3 of Venus Publication Question Bank

Q6. What is the difference between thermoplastics and thermosetting plastic ?

Ans. Refer to Chapter 3 of Venus Publication Question Bank

OR

Q6. Define a lubricant. With suitable examples, classify lubricant

Ans. Lubricants :

In order to reduce frictional resistance and wear and tear of machine parts, a layer of fluid is introduced in between the moving parts. This layer of fluid is known as lubricant. The lubricant prevents direct contact of the rubbing surfaces by forming a film in between. Lubricant also acts as a coolant by removing the frictional heat generated due to rubbing of the surfaces.

Classification of Lubricants :

According to the state of lubricants, these are mainly classified into the following classes :

1. Solid lubricants
2. Semi - solid lubricants
3. Liquid lubricants.

1. **Solid Lubricants :**

The lubricants existing in solid form are known as solid lubricants. e.g. graphite, molybdenum disulfide, soap, wax, mica, chalk, etc. These lubricants are used in heavy machinery which operates under very heavy load and low speed. Recently some stearates and palmitates of aluminium and mineral acid have also been employed as solid lubricants. Graphite is an expensive solid lubricant, but is widely used for solid lubrication. MoS_2 is also used as a solid lubricant.

2. **Semi - Solid Lubricants :**

These are neither solid nor liquid. A semi - solid lubricant is obtained by combining lubricating oil with thickening agent. Lubricating oil is the principal component and it can be either petroleum or a synthetic hydrocarbon of low or high viscosity. The thickeners consist primarily of special soaps of lithium, sodium, calcium barium, aluminium etc. Non - soap thickeners include carbon black, silica, gel polyureas and other synthetic polymers. These type of lubricants have higher frictional resistance than oils and therefore can support much heavier loads at lower speeds.

3. **Liquid Lubricants :**

These exist in liquid form and are used to reduce friction or wear between two sliding metallic surfaces by providing a uniform film between them. These are also very important and are widely used in several machines,

tools etc because they act as : 1. Sealing agent, 2. Corrosion inhibitors, and 3. Cooling medium.

Group - C

Q7. An electric current is passed through CuSO_4 and AgNO_3 solution connected in series. If at any time 0.7gm copper is deposited, then how much Ag will be deposited ? {Cu=63.5}.

Ans: Weight of Cu deposited = 0.7gm.

Weight of Ag deposited = x

Equivalent weight of Cu

$$= \frac{\text{At. wt}}{\text{valency}} = \frac{63.5}{2} = 31.75$$

Equivalent weight of (Ag)

$$= \frac{\text{At. wt}}{\text{valency}} = \frac{108}{1} = 108.$$

According to Faraday's second law of electrolysis

$$\frac{\text{wt. of silver deposited}}{\text{wt. of copper deposited}} = \frac{\text{Eq. wt. of silver}}{\text{Eq. wt. of copper}}$$

$$= \frac{x}{0.7} = \frac{108}{31.75}$$

$$= x = \frac{108 \times 0.7}{31.75} = 2.38 \text{ gm Ans.}$$

Hence, weight of Ag deposited = 2.38gm.

OR

Q7. Write down the name and formula of important ores of Aluminium. Describe a method for extraction of Aluminium method from its ore, Write their uses.

Ans. Refer to Chapter 3 of Venus Publication Question Bank

Q8. What is electrolysis? Write down different uses of electrolysis in details?

Ans: These are the substances which follow the passage of electricity through their molten state or through their aqueous solutions and also undergo chemical decomposition at the same time. Some common examples of electrolytes are acids, bases and salts. The condition through the electrolytes is due to the movement of ions.

Uses of electrolysis :

1. Electrolysis is used in the extraction of metals from their ores. For example, when a current is passed through molten sodium chloride, sodium is deposited at the cathode and chlorine gas is evolved at the anode. Aluminium and potassium are also extracted by electrolysis.
2. It is used for refining certain metals such as copper and zinc.

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3. Electrolysis is used for the manufacture of chlorine. In submarines, oxygen produced by the electrolysis of water is used for breathing.
4. Electrolysis is used for electroplating many things we use every day.

OR

Q8. In what respects do primary cells differ from the secondary cells? Write the cell reaction of a lead storage battery when it is discharged. How does the density of the electrolyte change when the battery is discharged?

Ans. Refer to Chapter 4 of Venus Publication Question Bank

Q9(a). Define degree of hardness? How the degree of hardness of water is expressed.

Ans. Degree of hardness: The degree of hardness is defined as the number of parts by weight of calcium carbonate hardness per particular number of part of water, depending upon the unit employed.

The units commonly used to express hardness of water are as follows:

- (1) Parts per million (ppm) is the parts of calcium carbonate equivalent hardness per million part of water.
 $1 \text{ ppm} = 1 \text{ part of } \text{CaCO}_3 \text{ equivalent hardness per million parts of water}$
- (2) Milligrams per litre (mg/litre) is the number of milligrams of CaCO_3 equivalent hardness present per litre of water.
 $1 \text{ mg/litre} = 1 \text{ mg of } \text{CaCO}_3 \text{ equivalent in 1 litre}$
 Now weight of 1 litre water = 1000 gm = $1000 \times 100 \text{ mg} = 10^6 \text{ mg}$.
 $1 \text{ mg/litre} = 1 \text{ mg of } \text{CaCO}_3 \text{ equivalent per } 10^6 \text{ mg of water}$
 $1 \text{ mg/litre} = 1 \text{ ppm}$
- (3) Clarke's method: Clarke's degree is the number of grains (1/7000 lb) of CaCO_3 equivalent per gram (10 lb) of water.
 i.e. one part of CaCO_3 equivalent per 70,000 parts of water.
 $1^\circ\text{Cl} = 1 \text{ part of } \text{CaCO}_3 \text{ equivalent per 70,000 parts of water}$
- (4) Degree french is the parts of CaCO_3 equivalent per 1,00,000 parts of water.
 $1^\circ\text{Fr} = 1 \text{ part of } \text{CaCO}_3 \text{ equivalent per 105 parts of water}$
 $1 \text{ ppm} = 1 \text{ mg/lit} = 0.1^\circ\text{Fr} = 0.07^\circ\text{Cl}$

Q9(b). Differentiate between temporary and permanent hardness?

Ans. Refer to Chapter 2 of Venus Publication Question Bank

OR

Q9. Write important ores of Iron. How cast iron is extracted from it and reactions involved in various steps of extraction. Write their uses.

Ans. Refer to Chapter 3 of Venus Publication Question Bank

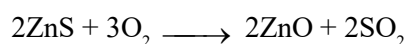
Q10. (a) Why carbonate and sulphide ores are usually converted into oxides during the process of extraction?
 (b) Write down electronic configuration of ferrous and cupric ions.

Ans. (a) In the extraction of metals from their ores, carbonate and sulfide ores are usually converted into oxides because:

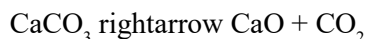
- (i) **Oxides are easier to reduce:** Most metals are extracted by reducing their oxides. The reduction of metal oxides to pure metals is relatively simple and more efficient compared to sulfides or carbonates. Oxides can be reduced either by heating with carbon (smelting) or by electrolysis, depending on the metal.
- (ii) **Stability of oxides:** Metal oxides are generally more stable than their sulfide or carbonate forms, which makes them suitable for reduction reactions. This stability helps in maintaining controlled reaction conditions during extraction.
- (iii) **Avoiding environmental issues:** When sulfide ores are roasted, sulfur dioxide (SO_2) is produced, which is a toxic gas and a major pollutant. Converting sulfides to oxides through controlled roasting minimizes sulfur pollution, as SO_2 can be captured and processed further. On the other hand, carbonates release carbon dioxide (CO_2) during conversion, which is less hazardous than SO_2 .
- (iv) **Easier thermal decomposition:** Carbonates and sulfides often require higher temperatures for direct extraction. Converting them into oxides through processes like calcination (for carbonates) and roasting (for sulfides) simplifies the subsequent reduction process, as oxides typically require lower temperatures for metal extraction.

For example:

Roasting of sulfide ores (e.g., zinc sulfide, lead sulfide) converts them into metal oxides and sulfur dioxide:



Calcination of carbonate ores (e.g., calcium carbonate) converts them into metal oxides and carbon dioxide:



Thus, converting ores into oxides streamlines the metal extraction process and minimizes environmental con-

cerns.

Ans.(b) Electronic configuration of Ferrous Fe^{+2} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.

Electronic configuration of cupric ions is $\{\text{Ar}\} 4s^0 3d^9$.

Q10.(b) Explain the following terms applied to lubricants.

- (a) Viscosity.
- (b) Viscosity index
- (c) Volatility
- (d) Oiliness.
- (e) Pour and cloud point.
- (f) Flash point and fire point.

Ans.(a) Viscosity : The resistance offered by liquids to flow is called as viscosity. OR The force in dynes required to move 1sq. cm of the liquid over another surface with a velocity of 1cm/sec. The unit of viscosity is poise. A high viscosity is thick and flows slowly. A low viscosity oil is thin and flows readily. viscosity is determined by using Redwood viscometer.

Ans. (b) Viscosity index : The rate of change of viscosity of liquid with temperature is known as viscosity index. The viscosity of liquid varies with the change of temperature. A good lubricant is that whose viscosity does not vary much or differ with the temperature.

Ans.(c) Volatility: Volatility of a liquid is its tendency to vaporize with the increase of temperature. If a lubricant is highly volatile, it will vaporize rapidly at low temperature. Hence extra quality of lubricant is required. The lubrication process will be costly. A good lubricant should have low volatility.

Ans.(d) Oiliness: It is defined as the power of an oil to maintain a continuous film under pressure while used as a lubricant. It is the property of an oil by which it sticks on the surface of machine parts working under high pressure and high temperature, Mineral oils have comparatively low oiliness than the vegetable oils. The oiliness of the mineral acid can be increased by the addition of fatty acids such as stearic acid or oleic acid.

Ans.(e) Pour and cloud point : Cloud point can be defined as the temperature at which the oil becomes cloudy or hazy in appearance (due to separation of wax). Pour point can be defined as the temperature at which oil ceases to flow or pour on cooling. These characteristics indicate the suitability of oils at lower temperature or in cold conditions. The lubricating oils used in machines working at low temperatures, should have much lower cloud and pour points than the working temperatures. Otherwise, the lubricating oil may get solidify at working temperature and this may cause jamming of the machine

parts. This may affect the speed of the machine.

Ans.(f) Flash point and fire point : Flash point defined as the minimum temperature at which the oil gives enough vapours which give momentary flash of light when a flame is brought near it. Fire point is defined as the minimum temperature at which oil produces enough vapours which catch fire and burn continuously at least for five seconds when a flame is brought near it. If the lubricant used is having low flash and fire points, then it may catch fire easily as the temperature increases due to the working conditions. Hence, it is preferable to use a lubricant having high flash and fire points.

OR

Q10. Write down the names and formula of different important ore of Nickel. How is nickel extracted from its ore?

Ans. Refer to Chapter 3 of Venus Publication Question Bank

Q11. Write short note on any two of the following-

- (a) Ohm's law
- (b) Specific conductivity
- (c) Metallurgy.
- (d) Functions of Lubricants

Ans.(a) Ohm's law : According to Ohm's the potential difference across a conductor is directly Proportional to the current flowing through it.

If V is the potential difference (in volts) across a conductor and I is the strength of the current (in amperes) flowing through it, then according to Ohm's law,

$$V \propto I$$

$$\text{or } \frac{V}{I} = \text{constant}$$

The constant of proportionality in the above equation is termed as resistance of the conductor and is represented as R. Therefore, we have

$$\frac{V}{I} = R$$

Eq. (i) represents the mathematical form of Ohm's law. The resistance of a conductor is measured in the unit Ohm's which is represented as Ω (omega).

Ohm's law can also be stated as follows. The strength of current flowing through a conductor is directly proportional to the potential difference across the conductor and inversely proportional to the resistance of the conductor.

Ans.(b) Specific conductivity : Refer to Chapter 5 of Venus Publication Question Bank

Ans.(c) Metallurgy : Refer to Chapter 3 of Venus Publication Question Bank

Ans.(d) Functions of Lubricants :

Lubricants perform various functions. Some of these

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are :

- Reduces the loss of energy in the form of heat.
- Increases the efficiency in the form of heat.
- Increases the efficiency of the machine.
- Prevents interlocking or inter joint welding at the surface asperities.
- Increases the smooth motion of moving parts.
- Reduces surface deformation, wear and tear.
- Reduces the expansion of metal by local frictional heat.
- Protects the material from corrosion.

OR

Q11. Write short note on any two of the following

- Aufbau principle
- Electro chemical series
- Viscosity and viscosity Index

Ans: (i) Aufbau Principle: Refer to Chapter 1 of Venus Publication Question Bank

(ii) Electro chemical series: The arrangement of metal ions in order of increasing values of standard reduction potential is known as electrochemical series.

Important features of electrochemical series :

- Metals near the top of the series are strongly electropositive (or weakly electronegative). They lose electrons readily to give cations.
- Lower the position of metal in the series, greater is its tendency to be reduced. Thus, metals having higher position in the series are strong reducing agents and their ions are stable. On the other hand, those near the bottom of the series are stable metals and their ions are easily reduced to the metals.
- Weakly electronegative metals in the upper part can displace more electronegative metals below them from their salts. For examples, iron displaces copper from CuSO_4 solution, Cu displaces silver from silver salts solution, silver displaces gold from gold salt solution as in photography during gold toning process.
- Metal above hydrogen displaces hydrogen from dilute acids.
- An oxidising agent with a higher reduction potential will oxidise any reducing agent with a lower reduction potential, i.e., silver (reducing potential + 0.80 V) will be deposited from its solution when treated with copper metal (reduction potential + 0.34 V).
- A reducing agent with a lower reduction potential will reduce any oxidising agent with a higher reduction potential, i.e., Li^+/Li has the lowest reduction potential. This means that all ions below it can be reduced by lithium.

Ans. (iii) Viscosity and viscosity Index :

Viscosity : Viscosity is defined as the property of a fluid which offers resistance to the movement of one layer of fluid over another adjacent layer of the fluid.

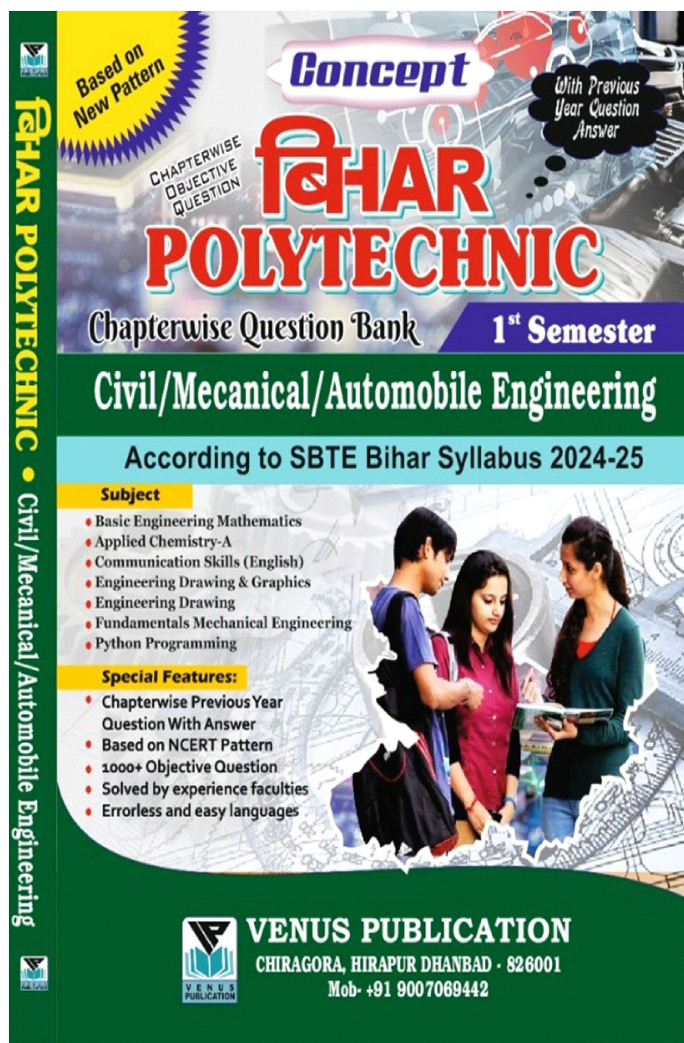
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Absolute viscosity: absolute viscosity, η is defined as the force per unit required to maintain the unit velocity difference between two parallel layers of fluid unit distance apart.

Relative viscosity: relative viscosity is the ratio of the solution. To the viscosity of the solvent used.

Viscosity index: viscosity index is defined as the rate at which viscosity of the lubricants oil changes with respect to temperature.

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